

Mid-Way Report

Submission for Feedback

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Dataset

We plan to use the [IMDB Movies Dataset](#), which contains comprehensive film data records, extracted from the IMDB database, which covers a range of titles and other related metadata. IMDB is one of the world's most authoritative and widely used sources for film, tv and entertainment data. The dataset includes features such as title IDs, release year, genres, runtime, ratings, cast, plot summaries and more. Many of these features are vast in analytical capability, they provide downstream tasks including, temporal trend analysis, network mapping amongst cast/directors, rating prediction.

After a brief overview of the dataset, it is visibly clear there will be a multitude of cleaning operations required, purely based upon simple logic which is good as it provides us with work to refine our 2.5 million row dataset down to a smaller (but still large), quality dataset.

Technology

Apache Pig - Pig is the ideal cleaning technology for handling csv files, identifying NULL or missing values and correctly identifying data types.

Apache Spark - For large scale data processing. Also can use SparkML for machine learning/predictive analytics.

Apache Hive - For handling SQL-like queries from the clean data outputted by Spark and/or Pig, stored in HDFS.

Matplotlib - For clear visualisations of plotted distributions and comparing values across a range of filtered columns.

Analytics

- Movie ratings by genre/director/actor and comparing ratings between genres/directors/actors.
- Gross income by genre/director/actor and comparing between genres/directors/actors.
- Count of movies/series by year (per genre/total) : trend of production over time.
- Time-series analysis: for each decade, what was the average rating, or how many high-rated movies (>8) were produced.
- Build a model to predict rating based on genre/runtime/year/country.
- Gap analysis: what genres are under-represented but have high ratings?

We want to analyse these aspects of the dataset to better understand what factors contribute to success in the film industry. By examining ratings and gross income across genres, directors and actors, we can identify patterns in audience preferences and financial success.

Examining movie counts and average ratings through years can help us observe trends over time and understand how the industry is changing.

Building a predictive model for ratings provides insight into the key features of how a movie performs/ is perceived and performing a gap analysis helps show under-represented genres that typically perform well, pointing to an opportunity for filmmakers to explore niche areas with strong audience appeal.

Team Roles and Tasks

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Resource Investigator - AD

Specialist - BC

Completer Finisher - BC

Plant - BC

Shaper - AD

Coordinator - LH

Monitor Evaluator - LH

Implementer - LH

Teamworker - AD

Specific Tasks

- Identifying errors within the dataset
- Run queries using Apache Hive, Pig and Spark
- Clean data through Apache Pig
- Compiled the final report
- Forked and updated gitlab